

Male Urethra

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Introduction

- Male urethra is 18–20 cm long
- extends from the internal urethral orifice in the urinary bladder to the external opening (meatus) at the end of the penis

Parts of Urethra

- Posterior Part
 - 4cm long
 - Divided into 3 parts
 - Prestatic
 - Prostatic
 - Membranous
- Anterior Part
 - 16cm long
 - Divided into 2 parts
 - Bulbar
 - Pendulous/Penile

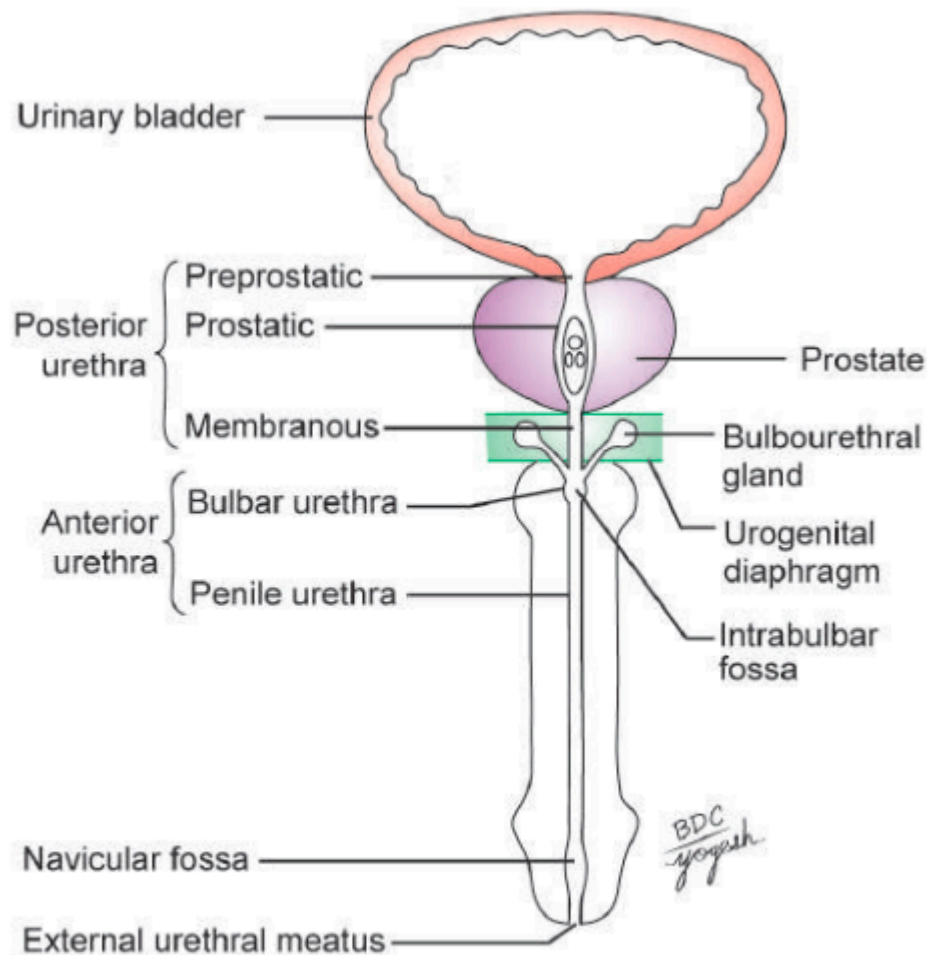


Fig. 30.10: Parts of male urethra (anterior view of the male urethra straightened and cut open)

Shape of Urethra

- In flaccid penis, urethra as a whole represents double curve except during the passage of fluid along it.
- The urethral canal is a mere slit.
- In transverse section:

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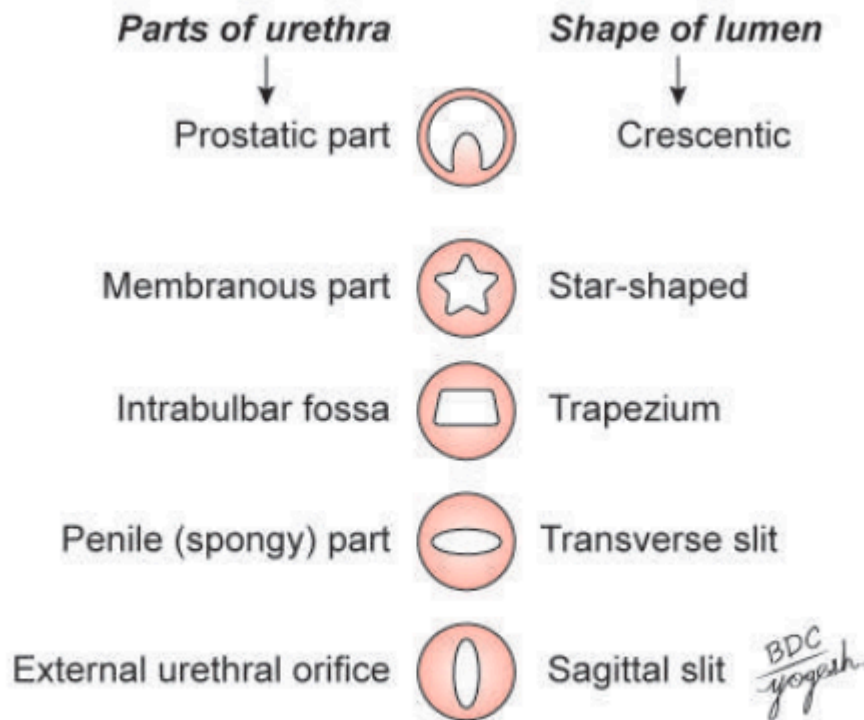


Fig. 30.11: Shape of different parts of the male urethra

Sphincters of Urethra

- Two sphincters: Internal & External urethral sphincters
- Difference b/w Internal & External urethral sphincters

◦ **TABLE 30.1: Differences between internal and external urethral sphincters**

<i>Internal urethral sphincter</i>	<i>External urethral sphincter</i>
Surrounds the internal urethral orifice	Surrounds the membranous urethra
Also called preprostatic sphincter in men or sphincter vesicae	Also called sphincter urethrae
Involuntary	Voluntary
Consists of smooth muscle fibres	Consists of striated muscle fibres
Innervated by sympathetic fibres (T11–L2)	Innervated by pudendal nerve (S2, 3, 4)
<i>Functions:</i> • Contenance of urine • Prevention of retrograde ejaculation	<i>Functions:</i> • Contenance of urine • Voluntary voiding of urine

Features of Urethra

- Posterior Urethra: 3 parts
 - Preprostatic urethra:
 - 1 cm long.
 - extends from the base of urinary bladder to the prostate.
 - surrounded by proximal urethral sphincter in males.
 - Prostatic urethra: (*make flowchart*)
 - 3–4 cm long.
 - widest and most dilatable part of male urethra
 - traverses through the substance of prostate closer to the anterior than the posterior surface of gland
 - Lowermost part of prostatic urethra is fixed by puboprostatic ligaments and is, therefore, immobile
 - Shows following features - (Urethral Crest, Prostatic Sinus, Verumontanum, Prostatic Utricle)
 - Urethral crest:
 - median longitudinal mucosal crest

- projects posteriorly into the lumen making the lumen crescentic in transverse section
- Prostatic Sinuses:
 - Depressions on each side of urethral crest
 - Shows 15-20 prostatic ducts
- Verumontanum/Seminal Colliculus:
 - Rounded elevation in middle of urethral crest
 - 3 orifices: Middle Prostatic utricle, 2 Ejaculatory Ducts
- Prostatic Utricle:
 - Cul-de-Sac (*a street closed at one end*)
 - consists of openings of numerous glands
 - Corresponds to Vagina (*vaginus masculinus*)
 - Develops from Mullerian Ducts

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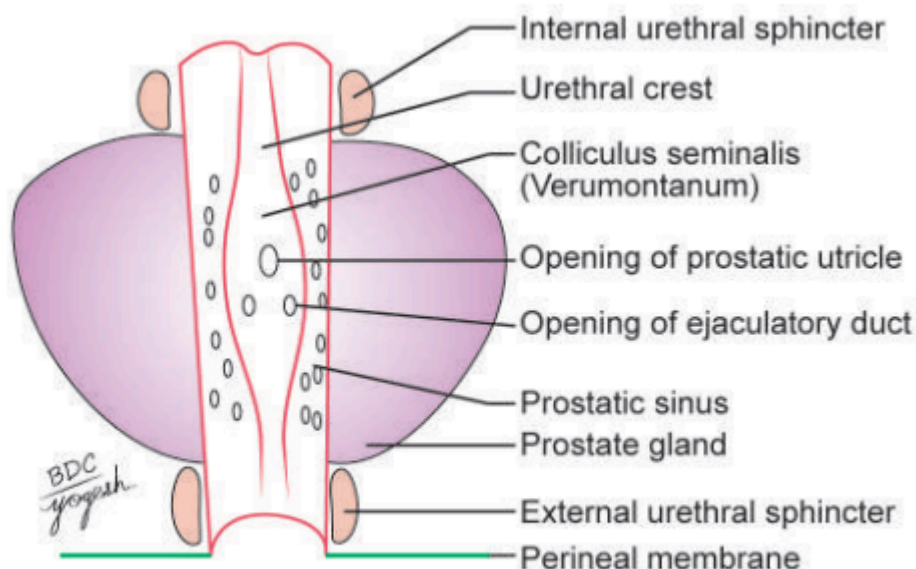


Fig. 30.12: Features of the posterior wall of the prostatic urethra (anterior view, coronal section)

- Membranous Urethra:
 - 2 to 2.5cm long
 - Second narrowest part of urethra (Ext. urethral orifice is the narrowest)
- Anterior Urethra: 2 parts
 - Bulbar Urethra: (entirely within perineum)
 - Surrounded by bulbospongiosus
 - has openings of bulbourethral glands
 - Penile/Pebdulous Urethra: (narrow, slit-like passage upto ext. urethral orifice)
 - Following features are seen - (Navicular fossa, External urethral orifice, Urethral glands of Littre, Urethral Lacuanae, Lacuna magna/sinus of guerin)
 - Navicular Fossa:
 - dilatation of terminal part of penile urethra within the glans penis

- External urethral orifice:
 - narrowest part of urethra.
 - a slit-like opening
- Urethral glands of Littre:
 - simple tubular mucous glands that open in the entire anterior urethra.
- Urethral lacunae:
 - pit-like recesses of urethral mucosa which project from entire urethra except in the navicular fossa.
- Lacuna magna/Sinus of Guérin:
 - the largest urethral lacuna located on the roof of the navicular fossa.
 - Its mouth is guarded by mucous fold called *valvule of Guérin*

Arteries

- i. Urethral artery arises from internal pudendal artery just below perineal membrane and travels through corpus spongiosum to reach glans penis.
- ii. Dorsal penile artery via circumflex branches on each side

Veins

- i. Anterior urethra → dorsal vein of penis → internal pudendal vein which drains into prostatic venous plexus → internal iliac veins.
- ii. Posterior urethra → prostatic and vesical venous plexus → internal iliac veins.

Lymphatic Drainage

- i. Prostatic urethra → internal iliac.
- ii. Membranous urethra → internal iliac.
- iii. Anterior urethra → accompany that of glans → deep inguinal.

Nerve Supply

- Prostatic plexus supplies the smooth muscle of prostate and prostatic urethra
- Parasympathetic nerves are from 2nd–4th sacral segments
- Greater cavernous nerves are sympathetic to preprostatic sphincter during ejaculation
- Nerve supply of rhabdosphincter is controversial but is said to be by neurons in Onuf's nucleus situated in 2nd sacral segment of spinal cord; fibres pass via perineal branch of pudendal nerve.

Clinical

- Urethritis:
 - an infection of urethra
- Hypospadias:
 - common anomaly in which the urethra opens on the undersurface of the penis or in the perineum
 - It may be:
 - glandular *on ventral surface of glans*
 - balanic *at base of glans*
 - penile *on ventral surface of penis*
 - penoscrotal *at junction of penis and scrotum*
 - perineal *at unfused part of scrotum*
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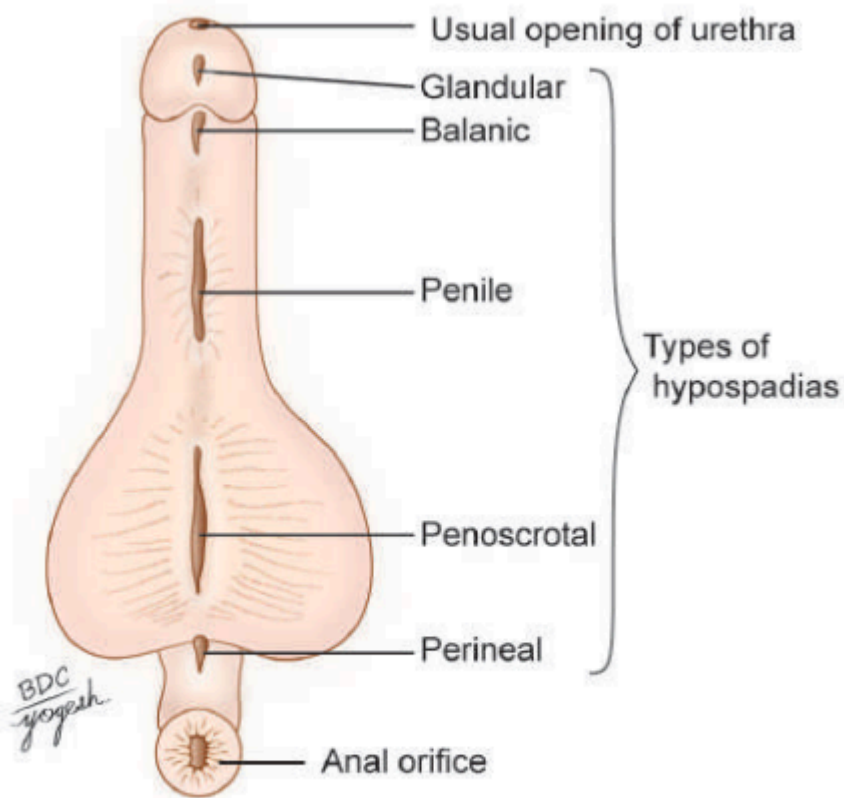


Fig. 30.14: Types of hypospadias

- Epispadias:
 - rare condition that causes failure of development of infraumbilical part of anterior abdominal wall
 - leads to opening of urethra or bladder on the dorsum of penis.
 - It may be associated with ectopia vesicae